The Mediating Role of Psychological Resilience in the Relationship Between Emotion Regulation and Psychological Well-being Among Caregivers of Children with Autism Spectrum Disorder

A Research Synopsis Submitted in Partial Fulfillment of the Requirements for the Degree of
Master of Philosophy (MPhil) in Psychology

Submitted by:
Syed Bilawal Jaffri

Department of Psychology
DHA Suffa University, Karachi
2026

Abstract

Caregivers of children diagnosed with Autism Spectrum Disorder (ASD) face chronic caregiving demands that place them at heightened risk of psychological distress and reduced well-being (Hayes & Watson, 2013; Nadeem et al., 2024). The capacity to regulate one's emotions has been identified as an important protective resource for parents of autistic children (Dimachkie Nunnally et al., 2023; Dominguez Ortega et al., 2026), yet the mechanism through which emotion regulation translates into well-being remains insufficiently understood, particularly in Pakistan. Drawing on Gross's (1998) process model of emotion regulation, Fredrickson's (2001) broaden-and-build theory, and Ryff's (1989) multidimensional model of psychological well-being, the present study aims to examine whether psychological resilience mediates the relationship between emotion regulation and psychological well-being among primary caregivers of children with ASD. A quantitative, cross-sectional correlational design will be employed. A purposive sample of approximately 150 to 200 primary caregivers, determined through a priori power analysis in G*Power 3.1 ($f^2 = .15$, $\alpha = .05$, power = .95), will be recruited from autism rehabilitation centres, special education institutes, and community mental health services in Hyderabad and Karachi,

Sindh, Pakistan. Emotion regulation will be assessed using the Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004), psychological resilience using the Brief Resilience Scale (Smith et al., 2008), and psychological well-being using Ryff's Scales of Psychological Well-being (Ryff, 1989). Data will be analysed using Pearson correlation, independent-samples *t*-tests, and simple mediation analysis via the PROCESS macro (Model 4) with 5,000 bootstrap resamples (Hayes, 2022). It is hypothesised that greater difficulties in emotion regulation will be associated with lower resilience and lower psychological well-being, and that resilience will significantly mediate this relationship. The findings are expected to identify resilience as a modifiable mechanism and inform caregiver-focused interventions in Pakistan.

Keywords: *Autism Spectrum Disorder, caregivers, emotion regulation, psychological resilience, psychological well-being, mediation*

Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterised by persistent deficits in social communication and social interaction, alongside restricted and repetitive patterns of behaviour, interests, or activities (American Psychiatric Association, 2013). Because the condition is lifelong and requires sustained support across multiple domains of functioning, its impact extends beyond the diagnosed child to the family system as a whole. The Autism Society of Pakistan has estimated that approximately 350,000 children in Pakistan are diagnosed with autism, with the true figure likely higher due to underdiagnosis and limited awareness among parents and healthcare providers (Akram et al., 2025).

Parenting a child with ASD involves demands that extend well beyond typical parenting responsibilities. A meta-analysis comparing parents of children with and without ASD found that

parents of children with ASD experienced substantially greater parenting stress, with a large effect size (Hayes & Watson, 2013). Pakistani evidence mirrors this pattern. In a study of 103 parents of children with ASD, 78% reported frequent caregiving-related stress and 60% scored above the clinical threshold on the Self-Reporting Questionnaire (SRQ-20), indicating a probable mental disorder (Nadeem et al., 2024). Similarly, a study of parents in Karachi found that 24.5% reported moderate and 19.0% severe anxiety, with a notable proportion at risk of parental burnout (Akram et al., 2025), and a study of 300 Pakistani parents of children with ASD found that caregiver frustration was significantly and negatively associated with psychological well-being (Shabbir et al., 2025).

Not all caregivers, however, respond to these demands in the same way. Individual differences in *emotion regulation*, defined as the processes by which individuals influence which emotions they have, when they have them, and how they experience and express them (Gross, 1998), shape how caregivers respond to the daily stressors of raising a child with ASD. Gratz and Roemer (2004) conceptualised difficulties in emotion regulation as involving nonacceptance of emotional responses, difficulty engaging in goal-directed behaviour when distressed, impulse control difficulties, lack of emotional awareness, limited access to regulation strategies, and lack of emotional clarity. Among parents of autistic children, stronger emotion regulation abilities have been associated with less caregiver strain, fewer depressive and anxiety symptoms, and better well-being (Dimachkie Nunnally et al., 2023; Dominguez Ortega et al., 2026).

Psychological resilience, defined most simply as the ability to bounce back or recover from stress (Smith et al., 2008), has emerged as a key protective factor for families of children with ASD. An integrative review of 22 studies concluded that parents of children with ASD who possess indicators of resilience are better able to manage the adversity associated with caregiving (Bekhet

et al., 2012), and a recent systematic review identified psychological factors, coping styles, and social support as key determinants of parental resilience in this population (Md Salim et al., 2026). Resilience also appears to function as a mechanism through which other psychological resources affect parental outcomes; for example, resilience and psychological flexibility fully mediated the relationship between mindfulness and parenting stress among 181 Chinese parents of autistic children (Feng et al., 2025).

Theoretical and empirical work suggests that emotion regulation may be one of the foundations on which resilience is built. Resilient individuals have been shown to use positive emotions to recover more quickly from negative emotional experiences (Tugade & Fredrickson, 2004), and emotion regulation has been described as a central protective process in the face of stress (Troy & Mauss, 2011). Among Spanish parents of children with ASD, emotion regulation and cognitive reappraisal emerged as significant predictors of resilience (Sarhani-Robles et al., 2025). In turn, resilience has been found to predict subjective well-being among parents of children with ASD in Hyderabad and Karachi (Samoo et al., 2025) and to mediate the relationship between life satisfaction and mental well-being among parents of autistic children in Multan, Pakistan (Kazmi et al., 2025).

Psychological well-being in this study will be conceptualised according to Ryff's (1989) eudaimonic model, comprising six dimensions: autonomy, environmental mastery, personal growth, positive relations with others, purpose in life, and self-acceptance. The Ryff scales have been translated into Urdu and cross-validated for use with Pakistani adults (Jibeen & Khalid, 2012), and have been used with Pakistani parents of children with ASD (Shabbir et al., 2025).

Despite these converging lines of evidence, no identified study has examined emotion regulation, psychological resilience, and psychological well-being together within a single

mediation model among caregivers of children with ASD in Pakistan. Understanding whether resilience explains the link between emotion regulation and well-being would clarify a mechanism that could be directly targeted by caregiver-support interventions. This study aims to address this gap.

Problem Statement

Primary caregivers of children with Autism Spectrum Disorder in Pakistan experience high levels of stress, anxiety, and reduced psychological well-being as a result of the chronic demands of caregiving (Nadeem et al., 2024; Akram et al., 2025; Shabbir et al., 2025). International research indicates that caregivers' emotion regulation abilities are associated with better mental health and well-being (Dimachkie Nunnally et al., 2023; Dominguez Ortega et al., 2026), and that psychological resilience protects parents against the adverse effects of caregiving stress (Bekhet et al., 2012; Feng et al., 2025). However, the pathway through which emotion regulation influences psychological well-being, and specifically whether psychological resilience mediates this relationship, has not been examined among caregivers of children with ASD in Pakistan. This study therefore seeks to examine the mediating role of psychological resilience in the relationship between emotion regulation and psychological well-being among primary caregivers of children with ASD.

Rationale and Significance of the Study

Conceptual Significance

This study integrates Gross's (1998) process model of emotion regulation, Fredrickson's (2001) broaden-and-build theory, and Ryff's (1989) model of psychological well-being into a

single explanatory model. Rather than examining direct associations only, it tests a mechanism (resilience) that may explain *how* emotion regulation relates to caregiver well-being, consistent with calls to investigate the processes underlying caregiver adaptation (Dominguez Ortega et al., 2026).

Population

Caregivers of children with ASD represent a population under chronic psychological strain, as documented in multiple Pakistani studies (Nadeem et al., 2024; Akram et al., 2025). Despite this, caregiver mental health remains a secondary consideration within most ASD-focused services in Pakistan, which predominantly target the diagnosed child rather than the family system.

Combination of Variables

Pakistani studies have examined resilience with subjective well-being (Samoo et al., 2025) and with life satisfaction and mental well-being (Kazmi et al., 2025), and psychological well-being with frustration (Shabbir et al., 2025). However, no identified study has examined emotion regulation, resilience, and Ryff's multidimensional psychological well-being together within a mediation framework in this population.

Methodology

The use of bootstrapped mediation analysis through the PROCESS macro (Hayes, 2022) provides a robust test of the indirect effect that does not rely on the normality assumptions of older approaches (Preacher & Hayes, 2008), allowing a precise estimate of the role of resilience as a mechanism.

Literature Gap

Existing research on emotion regulation and resilience among parents of children with ASD has been conducted mainly in the United States, China, and Europe (Feng et al., 2025; Dominguez Ortega et al., 2026; Md Salim et al., 2026), and a recent systematic review explicitly

called for culturally sensitive research on parental resilience (Md Salim et al., 2026). The present study contributes evidence from Sindh, Pakistan, a context in which family-based caregiving is the norm and formal support services remain limited.

Research Objectives

1. To examine the relationship between emotion regulation (difficulties in emotion regulation) and psychological well-being among primary caregivers of children with ASD.
2. To examine the relationship between emotion regulation and psychological resilience among primary caregivers of children with ASD.
3. To examine the relationship between psychological resilience and psychological well-being among primary caregivers of children with ASD.
4. To examine the mediating role of psychological resilience in the relationship between emotion regulation and psychological well-being among primary caregivers of children with ASD.
5. To examine differences between mothers and fathers in emotion regulation, psychological resilience, and psychological well-being.

Research Questions

1. Is there a significant relationship between difficulties in emotion regulation and psychological well-being among primary caregivers of children with ASD?
2. Is there a significant relationship between difficulties in emotion regulation and psychological resilience among primary caregivers of children with ASD?

3. Is there a significant relationship between psychological resilience and psychological well-being among primary caregivers of children with ASD?
4. Does psychological resilience mediate the relationship between difficulties in emotion regulation and psychological well-being among primary caregivers of children with ASD?
5. Do mothers and fathers differ significantly in emotion regulation, psychological resilience, and psychological well-being?

Definition of Key Terms

Autism Spectrum Disorder (ASD)

A neurodevelopmental disorder characterised by persistent deficits in social communication and social interaction, and restricted, repetitive patterns of behaviour, interests, or activities, as defined by the DSM-5 (American Psychiatric Association, 2013). For the purposes of this study, ASD will refer to a formal diagnosis made by a qualified medical or psychological professional.

Primary Caregiver

The individual who holds the primary responsibility for the daily care, supervision, and support of a child diagnosed with ASD, typically a parent or legal guardian.

Emotion Regulation

The processes by which individuals influence which emotions they have, when they have them, and how they experience and express these emotions (Gross, 1998). Difficulties in emotion regulation refer to nonacceptance of emotional responses, difficulties engaging in goal-directed behaviour, impulse control difficulties, lack of emotional awareness, limited access to emotion regulation strategies, and lack of emotional clarity (Gratz & Roemer, 2004).

Psychological Resilience

The ability to bounce back or recover from stress (Smith et al., 2008); more broadly, a process of positive adaptation in the context of significant adversity (Masten, 2001).

Psychological Well-being

A multidimensional construct comprising autonomy, environmental mastery, personal growth, positive relations with others, purpose in life, and self-acceptance (Ryff, 1989; Ryff & Keyes, 1995).

Mediation

A process in which the effect of a predictor variable (X) on an outcome variable (Y) is transmitted, wholly or in part, through an intervening variable (M), quantified as the indirect effect *ab* (Hayes, 2022).

Literature Review

Caregivers of Children with ASD: Burden and Psychological Impact

Caregivers of children with ASD face distinct and prolonged psychological demands. Hayes and Watson (2013), in a meta-analysis of studies using comparison groups, found that parents of children with ASD reported significantly greater parenting stress than parents of typically developing children, with a large effect size, and greater stress than parents of children with most other disabilities. In Pakistan, Nadeem et al. (2024) reported that among 103 parents of children with ASD, 78% experienced frequent stress, 91% reported stress related to core autism behaviours, and 60% scored at or above the clinical threshold on the SRQ-20, with parents reporting reduced psychological health and worse social relationships than population norms. Akram et al. (2025) found that among parents attending rehabilitation centres in Karachi, 24.5% reported moderate anxiety and 19.0% severe anxiety, while 18.0% were at risk of parental burnout.

These findings establish caregivers of children with ASD in Pakistan as a population carrying a substantial and largely unaddressed psychological burden.

Stress and coping theory conceptualises such caregiving demands as chronic stressors that may exceed a person's perceived coping resources (Lazarus & Folkman, 1984). Within this view, outcomes depend not only on the severity of the stressor but on the internal resources, such as emotion regulation and resilience, that caregivers bring to bear on it. This perspective directs attention toward modifiable personal resources that may explain why some caregivers maintain well-being despite comparable demands.

Emotion Regulation Among Caregivers of Children with ASD

Gross (1998) described emotion regulation as a set of processes through which individuals shape the occurrence, intensity, and expression of emotions, distinguishing antecedent-focused strategies, such as cognitive reappraisal, from response-focused strategies, such as expressive suppression. Habitual use of reappraisal has been associated with more positive emotion, better interpersonal functioning, and greater well-being, whereas habitual suppression has been associated with poorer outcomes (Gross & John, 2003). Gratz and Roemer (2004) extended this work by developing a multidimensional, clinically oriented conceptualisation of emotion dysregulation, operationalised through the Difficulties in Emotion Regulation Scale (DERS).

Among parents of autistic children, emotion regulation has been identified as an important protective resource. Dimachkie Nunnally et al. (2023) examined child characteristics and parent mental health outcomes and explored parents' emotion regulation abilities as a factor that may ameliorate caregiver strain, anxiety, and depression. In a dyadic study of 263 mother–father pairs, Dominguez Ortega et al. (2026) found that both mothers' and fathers' emotion regulation were associated with their own depression and anxiety symptoms, caregiver strain, and well-being,

concluding that emotion regulation is "an important source of resilience" for these parents and calling for research on the mechanisms linking emotion regulation to well-being. A scoping review of digital health interventions for parents of children with ASD similarly identified parental emotion regulation as a major intervention focus and recommended it as an outcome for future research (Ji et al., 2025).

Psychological Resilience Among Caregivers of Children with ASD

Resilience has been described as "ordinary magic", a common adaptive process rather than a rare trait (Masten, 2001). Smith et al. (2008) argued that most resilience measures assess resources that make resilience possible rather than resilience itself, and developed the Brief Resilience Scale (BRS) to assess the ability to bounce back from stress directly. Among families of children with ASD, Bekhet et al. (2012) reviewed 22 studies and identified indicators, risk factors, protective factors, and outcomes of resilience, concluding that resilient parents are better able to manage the adversity of caregiving. A more recent systematic review of 13 studies published between 2019 and 2025 identified sociodemographic factors, child characteristics, parenting well-being and distress, psychological factors and coping styles, and social support as determinants of parental resilience (Md Salim et al., 2026).

Resilience has also been examined as a mediator in this population. Feng et al. (2025) found that resilience and psychological flexibility completely mediated the relationship between mindfulness and parenting stress among parents of autistic children in China. Within Pakistan, Kazmi et al. (2025) used the BRS with 200 parents of autistic children in Multan and found that resilience played a mediating role between life satisfaction and mental well-being, while Samoo et al. (2025), using the Urdu version of the Connor-Davidson Resilience Scale with 130 parents in Hyderabad and Karachi, found that resilience significantly predicted subjective well-being.

Emotion Regulation and Resilience

Several theoretical and empirical accounts position emotion regulation as a foundation of resilience. Tugade and Fredrickson (2004) demonstrated that resilient individuals use positive emotions to recover more quickly from negative emotional arousal, and Troy and Mauss (2011) argued that emotion regulation, particularly cognitive reappraisal, functions as a protective factor that helps individuals maintain functioning under stress. In parents of children with ASD, Sarhani-Robles et al. (2025) found that emotion regulation and cognitive reappraisal were among the significant predictors of resilience in a sample of 799 parents. Together, this evidence suggests that caregivers who are better able to recognise, accept, and manage their emotions are likely to recover more readily from caregiving-related stress.

Psychological Well-being (Ryff's Model) Among ASD Caregivers in Pakistan

Ryff (1989) proposed a eudaimonic model of psychological well-being consisting of six dimensions: autonomy, environmental mastery, personal growth, positive relations with others, purpose in life, and self-acceptance, and Ryff and Keyes (1995) subsequently provided support for this six-factor structure in a national sample. Jibeen and Khalid (2012) cross-validated the Urdu translation of the 54-item Ryff scale with 308 Pakistani adults, reporting a Cronbach's alpha of .843 for the overall scale. Shabbir et al. (2025) used the Ryff scale with 300 Pakistani parents of children with ASD and found a significant negative correlation between frustration and psychological well-being, which was more pronounced in mothers than fathers, confirming both the relevance of the measure and the presence of well-being deficits in this population.

The Research Gap

Although emotion regulation has been linked to caregiver well-being (Dominguez Ortega et al., 2026) and resilience has been linked to well-being among Pakistani parents of children with

ASD (Kazmi et al., 2025; Samoo et al., 2025), these variables have not been examined together within a single mediation model in Pakistan. In addition, existing Pakistani studies have measured subjective or mental well-being rather than Ryff's multidimensional psychological well-being. The present study addresses these gaps by testing whether psychological resilience mediates the relationship between emotion regulation and psychological well-being among primary caregivers of children with ASD in Sindh, Pakistan.

Theoretical Framework

This study draws upon three complementary theoretical perspectives: Gross's (1998) process model of emotion regulation, Fredrickson's (2001) broaden-and-build theory of positive emotions, and Ryff's (1989) multidimensional model of psychological well-being, situated within Lazarus and Folkman's (1984) transactional model of stress and coping.

Gross's Process Model of Emotion Regulation (Predictor)

Gross (1998) proposed that emotions unfold over time through a sequence of situation, attention, appraisal, and response, and that individuals can regulate emotions at each stage, through situation selection, situation modification, attentional deployment, cognitive change, and response modulation. Gratz and Roemer (2004) built on this foundation to identify specific difficulties that impair adaptive regulation. In the caregiving context, a parent who can accept, understand, and flexibly manage distress arising from a child's behavioural outbursts is better positioned to continue goal-directed caregiving rather than becoming overwhelmed. In this study, difficulties in emotion regulation serve as the predictor variable.

Fredrickson's Broaden-and-Build Theory (Mechanism)

Fredrickson (2001) proposed that positive emotions broaden people's momentary thought–action repertoires and, over time, build enduring personal resources, including psychological resilience. Effective emotion regulation allows caregivers to down-regulate negative emotions and access positive emotions even under stress, and Tugade and Fredrickson (2004) showed that resilient individuals use positive emotions to recover from negative emotional experiences. This theory explains the first path of the model (emotion regulation → resilience): caregivers with fewer regulation difficulties are expected to build greater capacity to bounce back from stress.

Ryff's Model of Psychological Well-being (Outcome)

Ryff (1989) conceptualised well-being as optimal psychological functioning across six dimensions. Resilience is expected to support these dimensions: a caregiver who recovers quickly from setbacks is better able to maintain a sense of environmental mastery, sustain positive relationships, and preserve purpose in life despite ongoing demands. This explains the second path of the model (resilience → psychological well-being).

Integration: Stress and Coping Perspective

Lazarus and Folkman (1984) proposed that psychological outcomes of stress depend on the appraisal of the stressor and the coping resources available. Within this framework, raising a child with ASD constitutes a chronic stressor; emotion regulation represents a regulatory resource used during appraisal and coping; and resilience represents the resulting capacity to recover, which in turn sustains well-being. Accordingly, the study proposes that emotion regulation influences psychological well-being both directly and indirectly through psychological resilience, as illustrated in Figure 1.

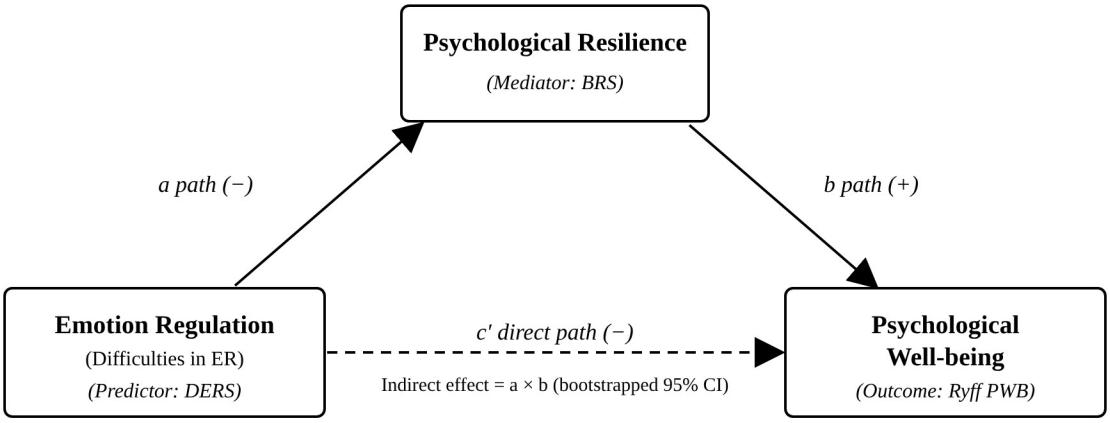


Figure 1. *Proposed conceptual model: psychological resilience as a mediator between emotion regulation and psychological well-being. Higher DERS scores indicate greater difficulties; hence negative a and c' paths are expected.*

Research Hypotheses

H1:
There will be a significant negative relationship between difficulties in emotion regulation and psychological well-being among primary caregivers of children with ASD.

H2:
There will be a significant negative relationship between difficulties in emotion regulation and psychological resilience among primary caregivers of children with ASD.

H3:
There will be a significant positive relationship between psychological resilience and psychological well-being among primary caregivers of children with ASD.

H4:
Psychological resilience will significantly mediate the relationship between difficulties in emotion regulation and psychological well-being among primary caregivers of children with ASD.

H5:

There will be significant differences between mothers and fathers in difficulties in emotion regulation, psychological resilience, and psychological well-being.

Design Study

Objectives

This study aims to examine the relationships among emotion regulation, psychological resilience, and psychological well-being, and to test the mediating role of psychological resilience, among primary caregivers of children with ASD, as detailed in the Research Objectives section above.

Hypotheses

It is hypothesised that difficulties in emotion regulation will be negatively associated with resilience and psychological well-being, that resilience will be positively associated with psychological well-being, and that resilience will mediate the relationship between emotion regulation and psychological well-being, as detailed in the Research Hypotheses section above.

Variables

Table 1. *Study variables and measures*

Role in Model	**Variable**	**Measure**
Independent / Predictor (X)	Emotion regulation (difficulties in emotion regulation)	Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)
Mediator (M)	Psychological resilience	Brief Resilience Scale (Smith et al., 2008)
Dependent / Outcome (Y)	Psychological well-being	Ryff's Scales of Psychological Well-being (Ryff, 1989)
Covariates / Demographics	Caregiver gender, age, education, family income, child's age and gender, time since diagnosis	Demographic Information Sheet

Operational Definitions

Difficulties in emotion regulation will be operationally defined as the total score (and six subscale scores) obtained on the Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004), with higher scores indicating greater difficulty. Psychological resilience will be operationally defined as the mean score obtained on the Brief Resilience Scale (Smith et al., 2008), with higher scores indicating greater resilience. Psychological well-being will be operationally defined as the total score and six subscale scores obtained on Ryff's Scales of Psychological Well-being (Ryff, 1989), with higher scores indicating greater well-being.

Strategy of the Research

This study will employ a quantitative, cross-sectional, correlational (mediational) survey design.

Method

Research Design

A quantitative, cross-sectional correlational design will be employed, in which all variables are measured at a single time point using standardised self-report questionnaires. This design is appropriate for examining naturally occurring relationships among psychological variables and for testing statistical mediation models, and has been used in comparable studies of resilience among parents of autistic children (Feng et al., 2025; Kazmi et al., 2025). The indirect (mediating) effect of resilience will be tested with the regression-based, bootstrapped approach described by Hayes (2022).

Participants

Who

Participants will be primary caregivers, typically parents, of children formally diagnosed with Autism Spectrum Disorder (American Psychiatric Association, 2013).

Size

The sample size was determined through an a priori power analysis using G*Power software, version 3.1.9.7 (Faul et al., 2007, 2009). Because the main analyses are regression-based (PROCESS Model 4 estimates the *b* and *c′* paths through a regression of psychological well-being on emotion regulation and resilience), the test family *F tests* and the statistical test *Linear multiple regression: Fixed model, R² deviation from zero* were selected. The effect size was set at $f^2 = .15$, which corresponds to a medium effect according to Cohen's (1988) conventions (small = .02, medium = .15, large = .35). A medium effect was chosen because previous studies with parents of children with ASD reported moderate associations among emotion regulation, resilience, and well-being (Feng et al., 2025; Samoo et al., 2025; Dominguez Ortega et al., 2026). The significance level was set at $\alpha = .05$ and statistical power $(1 - \beta)$ at .95, with two predictors. The input and output parameters are presented in Table 2.

Table 2. *G*Power a priori sample size calculation*

Parameter	**Value**
Software	G*Power 3.1.9.7 (Faul et al., 2009)
Test family	*F* tests
Statistical test	Linear multiple regression: Fixed model, R² deviation from zero
Type of power analysis	A priori: compute required sample size given α, power, and effect size
Effect size f^2	.15 (medium; Cohen, 1988)
α error probability	.05
Power $(1 - \beta)$	.95
Number of predictors	2 (emotion regulation, psychological resilience)
Output: Noncentrality parameter λ	16.05
Output: Critical *F*	3.08 (df = 2, 104)
Output: Actual power	.952
Output: Total sample size	**107**

As a supplementary check for the correlational hypotheses (H1–H3), G*Power (*Exact* test family, *Correlation: Bivariate normal model*, two-tailed) with a medium effect size of $r = .30$, α

= .05, and power = .95 indicated a required sample of 138. In addition, Fritz and MacKinnon (2007) recommend approximately 148 participants to detect an indirect effect with .80 power when both the *a* and *b* paths are small-to-medium (.26) using bias-corrected bootstrapping. Taking the largest of these estimates (148) and allowing for approximately 20% incomplete or invalid questionnaires, a target sample of approximately **150 to 200** caregivers will be recruited. This sample size is also comparable to recent studies of resilience among Pakistani parents of children with ASD (Samoo et al., 2025; Kazmi et al., 2025).

Where

Participants will be approached through autism rehabilitation centres, special education institutes, and community mental health services providing services to children with ASD and their families in Hyderabad and Karachi, Sindh, Pakistan.

Sampling Technique

Purposive sampling will be used. This non-probability sampling technique involves the deliberate selection of participants who meet predetermined criteria relevant to the research question, in this case primary caregivers of children with a formal ASD diagnosis. This approach is consistent with prior research with Pakistani parents of children with ASD (Shabbir et al., 2025; Samoo et al., 2025).

Specifications

Participants will be adults aged 18 years and above, of any gender, who are literate in Urdu or English.

Inclusion Criteria

- Primary caregiver (parent or legal guardian) of a child aged 3 to 18 years with a formal diagnosis of Autism Spectrum Disorder
- Aged 18 years or above

- Providing care to the child for at least six months since diagnosis
- Literate in Urdu or English

Exclusion Criteria

- Caregivers with a self-reported current, diagnosed severe psychiatric or neurological disorder
- Caregivers of a child whose ASD is accompanied by a severe physical disability or chronic medical illness, to avoid confounding caregiving demands
- Paid or professional caregivers (e.g., hired attendants or therapists)
- Caregivers who have experienced a major bereavement or traumatic life event within the past six months

Measures

Informed Consent Form

Prior to participation, all caregivers will be provided with an informed consent form outlining the purpose of the study, procedures, confidentiality protections, and their right to withdraw at any time without penalty, consistent with the procedures described by Shabbir et al. (2025).

Demographic Information Sheet

A demographic information sheet developed for this study will collect caregiver age, gender, relationship to the child, marital status, education level, employment status, monthly family income, family system (nuclear/joint), number of children, and the age, gender, and time since diagnosis of the child with ASD.

Difficulties in Emotion Regulation Scale (DERS)

What it measures: Difficulties in emotion regulation across six dimensions: nonacceptance of emotional responses, difficulties engaging in goal-directed behaviour, impulse control difficulties,

lack of emotional awareness, limited access to emotion regulation strategies, and lack of emotional clarity (Gratz & Roemer, 2004)

Author: Kim L. Gratz and Lizabeth Roemer

Year: 2004

Version: Original 36-item English version. Permission for use, and confirmation regarding a validated Urdu translation, will be sought prior to data collection; if no validated translation is available, a forward–backward translation procedure (Brislin, 1970) will be conducted with appropriate permissions

Number of Items: 36 items

Subscales: Nonacceptance (items 11, 12, 21, 23, 25, 29); Goals (items 13, 18, 20, 26, 33); Impulse (items 3, 14, 19, 24, 27, 32); Awareness (items 2, 6, 8, 10, 17, 34); Strategies (items 15, 16, 22, 28, 30, 31, 35, 36); Clarity (items 1, 4, 5, 7, 9)

Response Set: 5-point Likert-type scale ranging from 1 (*Almost Never*, 0–10%) to 5 (*Almost Always*, 91–100%)

Scoring: Items 1, 2, 6, 7, 8, 10, 17, 20, 22, 24, and 34 are reverse-scored. The total score is the sum of all 36 items (range 36–180); subscale scores are the sum of their respective items

Interpretation: Higher total and subscale scores indicate greater difficulties in emotion regulation

Psychometric Properties: In the original validation with 375 undergraduates, the DERS showed high internal consistency ($\alpha = .93$), good test–retest reliability over 4–8 weeks ($\rho = .88$), and adequate construct and predictive validity (Gratz & Roemer, 2004)

Brief Resilience Scale (BRS)

What it measures: Psychological resilience, defined as the ability to bounce back or recover from stress (Smith et al., 2008)

Author: Bruce W. Smith, Jeanne Dalen, Kathryn Wiggins, Erin Tooley, Paulette Christopher, and Jennifer Bernard

Year: 2008

Version: Original 6-item English version; Urdu translation via forward–backward translation (Brislin, 1970) with author permission, unless a validated Urdu version is located. The BRS has previously been used with Pakistani parents of children with ASD (Kazmi et al., 2025)

Number of Items: 6 items

Subscales: Unidimensional (three positively worded and three negatively worded items)

Response Set: 5-point Likert-type scale ranging from 1 (*Strongly Disagree*) to 5 (*Strongly Agree*)

Scoring: Items 2, 4, and 6 are reverse-scored. The total score is the mean of the six items (range 1–5)

Interpretation: Higher scores indicate greater resilience. Scores of 1.00–2.99 indicate low, 3.00–4.30 normal, and 4.31–5.00 high resilience (Smith et al., 2013)

Psychometric Properties: Across four samples, Cronbach's alpha ranged from .80 to .91, and test–retest reliability was .69 at one month and .62 at three months; the BRS showed good convergent and discriminant validity (Smith et al., 2008)

Ryff's Scales of Psychological Well-being

What it measures: Psychological well-being across six dimensions: autonomy, environmental mastery, personal growth, positive relations with others, purpose in life, and self-acceptance (Ryff, 1989)

Author: Carol D. Ryff

Year: 1989 (original development)

Version: The 54-item version (nine items per dimension); the Urdu translation cross-validated for Pakistani populations will be used (Jibeen & Khalid, 2012). This version has also been used with Pakistani parents of children with ASD (Shabbir et al., 2025)

Number of Items: 54 items

Subscales: Six subscales of nine items each: Autonomy, Environmental Mastery, Personal Growth, Positive Relations with Others, Purpose in Life, and Self-Acceptance

Response Set: 6-point Likert-type scale ranging from 1 (*Strongly Disagree*) to 6 (*Strongly Agree*)

Scoring: Negatively worded items (marked "rs" in the scoring key) are reverse-scored. Subscale scores are summed (range 9–54 each), and the overall score is the sum of all subscales (range 54–324)

Interpretation: Higher scores indicate greater psychological well-being, both at the subscale level and overall

Psychometric Properties: The Urdu translation demonstrated a Cronbach's alpha of .843 for the overall scale among Pakistani adults, although subscale-level internal consistency was modest (Jibeen & Khalid, 2012)

Suggested Applicable Scales

The three instruments provided (DERS, BRS, and Ryff PWB) are directly applicable to the variables of the present study and are recommended as the primary measures. The following alternatives are suggested should a shorter battery or a previously validated Urdu version be preferred, for example to reduce respondent burden for caregivers with limited time.

Table 3. *Suggested applicable scales*

Variable	**Recommended (Primary)**	**Alternative**	**Rationale for Alternative**
Emotion regulation	DERS, 36 items (Gratz & Roemer, 2004)	DERS-16 (Bjureberg et al., 2016)	16 items; retains the total score with good reliability and validity; reduces completion time
Psychological resilience	Brief Resilience Scale, 6 items (Smith et al., 2008)	Connor-Davidson Resilience Scale, CD-RISC-25 (Connor & Davidson, 2003)	Urdu version already used with parents of children with ASD in Hyderabad and Karachi (Samoo et al., 2025); measures resilience resources rather than recovery
Psychological well-being	Ryff PWB, 54 items, Urdu (Jibeen & Khalid, 2012)	Ryff PWB, 18-item short form (Ryff & Keyes, 1995)	Shorter; suitable when total battery length is a concern, although subscale reliability is lower

Note. The total battery using the primary measures is 96 items (36 + 6 + 54) plus the demographic sheet, requiring approximately 25 to 35 minutes. Using the DERS-16 instead of the full DERS would reduce the battery to 76 items.

Procedure

Permission

Formal ethical approval will be obtained from the Ethics Review Committee of DHA Suffa University prior to data collection. Institutional permission will also be obtained from the administration of each rehabilitation centre, special education institute, or mental health service where data collection will take place, consistent with the procedures described by Shabbir et al. (2025).

Permission of the Scale

Written permission to use the Difficulties in Emotion Regulation Scale, the Brief Resilience Scale, and the Urdu version of Ryff's Scales of Psychological Well-being will be requested from the respective scale authors, translators, or copyright holders, in accordance with standard academic research practice.

Translation (if required)

For any scale without a validated Urdu version, the forward–backward translation method (Brislin, 1970) will be used: two bilingual experts will independently translate the scale into Urdu, two different experts will back-translate it into English, and a committee will reconcile discrepancies. The translated version will be pilot tested with 15 to 20 caregivers to check comprehension and preliminary reliability; pilot participants will not be included in the main sample.

How Participants Will Be Contacted

Eligible caregivers will be identified through the registers and referral networks of the participating institutions. Staff will inform caregivers about the study, and those meeting the inclusion criteria will be invited to participate voluntarily, typically while waiting during their child's therapy sessions.

Order of Administration

Following informed consent, participants will complete the Demographic Information Sheet, followed by the Difficulties in Emotion Regulation Scale, the Brief Resilience Scale, and Ryff's Scales of Psychological Well-being. Questionnaires will be administered individually in a quiet room, and the researcher will be present to clarify instructions or read items aloud for caregivers who request assistance.

Time Taken

Completion of the full battery is estimated at approximately 25 to 35 minutes per participant. Total data collection is expected to take approximately three months.

Debrief

Upon completion, participants will be thanked and debriefed regarding the purpose of the study, and will be provided with information about psychological support services available through the host institutions should they wish to access further support.

Statistical Analysis

Data will be analysed using IBM SPSS Statistics (version 27 or later) with the PROCESS macro (version 4) (Hayes, 2022). The analysis will proceed as follows:

1. Data screening for missing values, outliers, and normality (skewness and kurtosis within ±2), and computation of descriptive statistics for demographic and study variables.
2. Internal consistency (Cronbach's alpha) of each scale and subscale in the present sample.
3. Harman's single-factor test to assess common method variance, as all variables are self-reported (Podsakoff et al., 2003).
4. Pearson product-moment correlations to test H1, H2, and H3.
5. Simple mediation analysis using PROCESS Model 4, with difficulties in emotion regulation as X, resilience as M, and psychological well-being as Y, to test H4. The indirect effect (*ab*) will be estimated with 5,000 bootstrap resamples, and mediation will be supported if the 95% bootstrap confidence interval does not include zero (Preacher & Hayes, 2008; Hayes, 2022). Relevant demographic variables (e.g., caregiver gender and child age) will be entered as covariates where they correlate significantly with the study variables. Supplementary analyses will repeat the model for each of the six Ryff subscales.

6. Independent-samples *t*-tests (with Cohen's *d*) to test H5, comparing mothers and fathers on all study variables.

Ethical Considerations

- Informed consent will be obtained from all participants prior to data collection, and participants will be informed of their right to withdraw at any point without any consequence to the services their child receives.
- All data will be kept confidential and stored securely; questionnaires will be coded numerically, and no identifying information will be reported.
- Participation will be entirely voluntary, with no coercion exerted by institutional staff or the researcher.
- Should any participant experience psychological distress while completing the questionnaires, they will be offered a break and referred to the appropriate clinical staff at the host institution.
- The study will be conducted in accordance with the ethical guidelines of the Pakistan Psychological Association and the ethical review requirements of DHA Suffa University.
- Permissions for the use and translation of standardised instruments will be obtained prior to data collection, in accordance with copyright and ethical research practice.

Implications

Field

This study contributes to the limited body of Pakistani research on caregivers of children with ASD by identifying whether psychological resilience is a mechanism through which emotion

regulation relates to psychological well-being, responding to calls for research on the processes underlying parental adaptation (Dominguez Ortega et al., 2026; Md Salim et al., 2026).

Psychologists

If resilience is found to mediate the relationship, psychologists can design caregiver interventions that target both emotion regulation skills (e.g., emotional awareness, acceptance, and reappraisal) and resilience-building, providing a theory-informed basis for intervention.

Professionals

Special educators, speech and language therapists, and behavioural therapists may benefit from increased awareness of caregiver emotional needs and may incorporate screening for emotion regulation difficulties and low resilience into routine family contact.

Policy Makers

The findings may support the inclusion of caregiver-focused psychological support services within autism rehabilitation and disability service frameworks in Pakistan.

Sustainable Development Goals (SDGs)

This research aligns with Sustainable Development Goal 3 (Good Health and Well-being), which emphasises the promotion of mental health and well-being for all, including caregivers within families affected by disability.

Media

The findings may raise public awareness of the psychological needs of caregivers of children with ASD and the role of emotional coping and resilience in sustaining family well-being.

Limitations

- The cross-sectional design does not permit causal conclusions; the hypothesised direction of effects is based on theory, and longitudinal research would be needed to confirm temporal ordering.
- Reliance on self-report measures may introduce social desirability and common method bias.
- Purposive sampling from institutions in Hyderabad and Karachi may limit generalisability to caregivers in rural areas or those not accessing services.

References

Akram, M., Naqvi, S. M. Z. H., & Jameel, N. (2025). Relationship between children's autism spectrum disorder and parental anxiety and burnout. *Pakistan Journal of Medical Sciences, 41*(2). https://doi.org/10.12669/pjms.41.2.9979

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). American Psychiatric Association. https://doi.org/10.1176/appi.books.9780890425596

Bekhet, A. K., Johnson, N. L., & Zauszniewski, J. A. (2012). Resilience in family members of persons with autism spectrum disorder: A review of the literature. *Issues in Mental Health Nursing, 33*(10), 650–656. https://doi.org/10.3109/01612840.2012.671441

Bjureberg, J., Ljótsson, B., Tull, M. T., Hedman, E., Sahlin, H., Lundh, L.-G., Bjärehed, J., DiLillo, D., Messman-Moore, T., Gumpert, C. H., & Gratz, K. L. (2016). Development and validation of a brief version of the Difficulties in Emotion Regulation Scale: The DERS-16. *Journal of Psychopathology and Behavioral Assessment, 38*(2), 284–296. https://doi.org/10.1007/s10862-015-9514-x

Brislin, R. W. (1970). Back-translation for cross-cultural research. *Journal of Cross-Cultural Psychology, 1*(3), 185–216. https://doi.org/10.1177/135910457000100301

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.

Connor, K. M., & Davidson, J. R. T. (2003). Development of a new resilience scale: The Connor-Davidson Resilience Scale (CD-RISC). *Depression and Anxiety, 18*(2), 76–82. https://doi.org/10.1002/da.10113

Dimachkie Nunnally, A., Factor, R. S., Sturm, A., Valluripalli Soorya, L., Wainer, A., Taylor, S., Ponzini, M., Abbeduto, L., & Gulsrud, A. C. (2023). Examining indicators of psychosocial risk and resilience in parents of autistic children. *Frontiers in Behavioral Neuroscience, 17,* 1102516. https://doi.org/10.3389/fnbeh.2023.1102516

Dominguez Ortega, L., Sturm, A., Krushena, M. M., & Gulsrud, A. C. (2026). Emotion regulation and self-inhibition's association with mental health outcomes, caregiver strain, and well-being in parents of autistic children: A dyadic analysis. *Journal of Neurodevelopmental Disorders, 18*. https://doi.org/10.1186/s11689-026-09704-1

Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*(2), 175–191. https://doi.org/10.3758/BF03193146

Faul, F., Erdfelder, E., Buchner, A., & Lang, A.-G. (2009). Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods, 41*(4), 1149–1160. https://doi.org/10.3758/BRM.41.4.1149

Feng, Y., Lu, D., Ng, T. Y. C., Chau, P. H., Chan, L. M. L., & Kwok, J. Y. Y. (2025). Mindfulness and parenting stress among parents of autistic children: The mediation of resilience and psychological flexibility. *Autism, 29*(7), 1849–1863. https://doi.org/10.1177/13623613251328465

Fredrickson, B. L. (2001). The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions. *American Psychologist, 56*(3), 218–226. https://doi.org/10.1037/0003-066X.56.3.218

Fritz, M. S., & MacKinnon, D. P. (2007). Required sample size to detect the mediated effect. *Psychological Science, 18*(3), 233–239. https://doi.org/10.1111/j.1467-9280.2007.01882.x

Gratz, K. L., & Roemer, L. (2004). Multidimensional assessment of emotion regulation and dysregulation: Development, factor structure, and initial validation of the Difficulties in Emotion Regulation Scale. *Journal of Psychopathology and Behavioral Assessment, 26*(1), 41–54. https://doi.org/10.1023/B:JOBA.0000007455.08539.94

Gross, J. J. (1998). The emerging field of emotion regulation: An integrative review. *Review of General Psychology, 2*(3), 271–299. https://doi.org/10.1037/1089-2680.2.3.271

Gross, J. J., & John, O. P. (2003). Individual differences in two emotion regulation processes: Implications for affect, relationships, and well-being. *Journal of Personality and Social Psychology, 85*(2), 348–362. https://doi.org/10.1037/0022-3514.85.2.348

Hayes, A. F. (2022). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach* (3rd ed.). Guilford Press.

Hayes, S. A., & Watson, S. L. (2013). The impact of parenting stress: A meta-analysis of studies comparing the experience of parenting stress in parents of children with and without autism spectrum disorder. *Journal of Autism and Developmental Disorders, 43*(3), 629–642. https://doi.org/10.1007/s10803-012-1604-y

Ji, B., Batubara, I. M. S., Batten, J., Peng, X., Chen, S., & Ni, Z. (2025). Digital health interventions targeting psychological health in parents of children with autism spectrum disorder: A scoping review. *BMC Psychology, 13*, 1128. https://doi.org/10.1186/s40359-025-03219-5

Jibeen, T., & Khalid, R. (2012). Cross validation of Ryff's scales of psychological well-being: Translation into Urdu language. *The International Journal of Educational and Psychological Assessment, 10*(2), 67–91.

Kazmi, S. F., et al. (2025). Mediating role of resilience in life satisfaction and mental wellbeing: Parental study with at least one child diagnosed with ASD. *The Critical Review of Social Sciences Studies*.

Lazarus, R. S., & Folkman, S. (1984). *Stress, appraisal, and coping*. Springer.

Masten, A. S. (2001). Ordinary magic: Resilience processes in development. *American Psychologist, 56*(3), 227–238. https://doi.org/10.1037/0003-066X.56.3.227

Md Salim, N. S., Idris, I. B., Nor, N. K., Roslan, S., Abdul Rashid, N., Ibrahim, R., & Abd Rahman, F. N. (2026). Factors associated with resilience among parents of children with autism spectrum disorder: A systematic review. *PLOS ONE, 21*(6), e0351969. https://doi.org/10.1371/journal.pone.0351969

Nadeem, N. J., Afzal, S., Dogar, I. A., Smith, J. G., Shah, T., Noreen, M., & Ali, A. (2024). The impact of having an autistic child on parental mental health and wellbeing in Pakistan. *Research in Autism Spectrum Disorders, 115*, 102423. https://doi.org/10.1016/j.rasd.2024.102423

Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879

Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods, 40*(3), 879–891. https://doi.org/10.3758/BRM.40.3.879

Ryff, C. D. (1989). Happiness is everything, or is it? Explorations on the meaning of psychological well-being. *Journal of Personality and Social Psychology, 57*(6), 1069–1081. https://doi.org/10.1037/0022-3514.57.6.1069

Ryff, C. D., & Keyes, C. L. M. (1995). The structure of psychological well-being revisited. *Journal of Personality and Social Psychology, 69*(4), 719–727. https://doi.org/10.1037/0022-3514.69.4.719

Samoo, M., et al. (2025). Resilience and subjective well-being: A study on parents of autism spectrum disorder children. *Journal of Social Research Development*.

Sarhani-Robles, A., Guillot-Valdés, M., Robles-Bello, M. A., & Sánchez-Teruel, D. (2025). Cognitive and emotional resilience in parents with children with autism spectrum

disorder during COVID-19: The role of promoting variables. *Journal of Intelligence, 13*(1), 6. https://doi.org/10.3390/jintelligence13010006

Shabbir, M., Arshad, S., Raza, M., Tahir, M., & Iqbal, Z. (2025). Frustration and psychological wellbeing among parents of children with autism. *Health and Research Insights, 3*(4), 58–62.

Smith, B. W., Dalen, J., Wiggins, K., Tooley, E., Christopher, P., & Bernard, J. (2008). The Brief Resilience Scale: Assessing the ability to bounce back. *International Journal of Behavioral Medicine, 15*(3), 194–200. https://doi.org/10.1080/10705500802222972

Smith, B. W., Epstein, E. M., Ortiz, J. A., Christopher, P. J., & Tooley, E. M. (2013). The foundations of resilience: What are the critical resources for bouncing back from stress? In S. Prince-Embury & D. H. Saklofske (Eds.), *Resilience in children, adolescents, and adults: Translating research into practice* (pp. 167–187). Springer.

Troy, A. S., & Mauss, I. B. (2011). Resilience in the face of stress: Emotion regulation as a protective factor. In S. M. Southwick, B. T. Litz, D. Charney, & M. J. Friedman (Eds.), *Resilience and mental health: Challenges across the lifespan* (pp. 30–44). Cambridge University Press.

Tugade, M. M., & Fredrickson, B. L. (2004). Resilient individuals use positive emotions to bounce back from negative emotional experiences. *Journal of Personality and Social Psychology, 86*(2), 320–333. https://doi.org/10.1037/0022-3514.86.2.320

Appendix

The following materials will be developed and attached as appendices in the final synopsis submission:

- Appendix A: Informed Consent Form
- Appendix B: Demographic Information Sheet
- Appendix C: Difficulties in Emotion Regulation Scale (DERS; 36 items) — to be attached upon receipt of permission
- Appendix D: Brief Resilience Scale (BRS; 6 items) — to be attached upon receipt of permission
- Appendix E: Ryff's Scales of Psychological Well-being (Urdu, 54 items) — to be attached upon receipt of permission
- Appendix F: Permission letters (university ethics committee, host institutions, scale authors)